

1. Administrative & Study Identification

Full Study Title: A Modular Phase I/II, Open-label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Antitumor Activity of AZD9829 as Monotherapy or in Combination in Patients With CD123-Positive Hematological Malignancies **Short Title/Acronym:** Study of AZD9829 in CD123+ Hematological Malignancies **Protocol Number:** D9470C00001 **NCT Number:** NCT06179511 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD9829 (CD123-directed antibody conjugated to a topoisomerase 1 inhibitor) **Phase of Development:** Phase I/II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the safety, tolerability, pharmacokinetics (PK), maximum tolerated dose (MTD), and Recommended Phase 2 Dose (RP2D) of AZD9829 as monotherapy or in combination in patients with CD123-positive hematological malignancies. * **Secondary Objectives:** To evaluate preliminary antitumor activity, progression-free survival (PFS), overall survival (OS), and immunogenicity of AZD9829.

2.2 Background & Rationale CD123 (interleukin-3 receptor alpha chain) is highly expressed on leukemic blasts and leukemia stem cells in diseases such as acute myeloid leukemia (AML) and myelodysplastic syndromes (MDS), while having limited expression on normal hematopoietic stem cells. This makes it an attractive therapeutic target. AZD9829 is a novel antibody-drug conjugate (ADC) designed to deliver a topoisomerase 1 inhibitor directly to CD123+ cells. This study addresses the urgent unmet medical need for patients with relapsed or refractory (R/R) AML and high-risk MDS who have exhausted standard treatment options.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Modular dose-setting study) * **Control Type:** Single-arm (Open-label dose escalation/expansion) * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target \$\$\$:** 56 participants (Actual enrollment) * **Number of Sites:** 22 locations * **Estimated Study Start:** 31-Jan-2024 (Actual) * **Estimated Primary Completion:** 31-Oct-2025 (Actual) / Study Completion: 23-Mar-2026 (Estimated)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Documented CD123+ hematologic malignancy based on flow cytometry or immunohistochemistry by local laboratory. 3. Relapsed/refractory (R/R) AML (except acute promyelocytic leukemia) OR R/R Higher-Risk MDS (HR-MDS) with $\geq 5\%$ bone marrow blasts at the time of inclusion. 4. Had at least 1 prior line of therapy at current histology and no available standard treatment options. 5. ECOG performance status of ≤ 2 .

4.2 Exclusion Criteria 1. Active central nervous system (CNS) leukemia. 2. Previous treatment with any CD123-targeting therapy. 3. Prior allogeneic hematopoietic stem cell transplantation (HSCT) within 90 days or cell therapy within 60 days of the start of therapy. 4. Active graft-versus-host disease (GVHD) requiring immunosuppressive treatment within 4 weeks prior to the start of AZD9829. 5. Active and uncontrolled infections or unresolved adverse events $\geq$ Grade 2 from prior therapies.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Ascending dose levels in dose escalation module, followed by dose optimization modules for AZD9829. * **Route of Administration:** Intravenous (IV) infusion. * **Duration of Treatment:** Until disease progression, unacceptable toxicity, or up to approximately 1 year.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard of care supportive therapies as required for the management of hematological malignancies and drug-related toxicities. * **Prohibited:** Other concurrent investigational anti-leukemic therapies or prior CD123-targeted agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, CD123 status verification, Bone Marrow biopsy, Baseline Labs
Baseline	Day 1	First IV Dose of AZD9829, PK sampling
Interim	Cycle 1 (28 days)	DLT Evaluation, PK/PD, Safety Labs, Efficacy Assessment
End of Study	Follow-up	Survival/PFS follow-up up to ~1 year; safety follow-up up to 60 days post-last dose

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Incidence and frequency of Dose-Limiting Toxicities (DLTs) to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). *

Measurement Method: Clinical and safety laboratory assessments evaluated during the 28-day Module 1 period.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetics: Area under the plasma concentration-time curve (AUC) and Maximum plasma concentration (C_{max}). * Progression-Free Survival (PFS): Time from the start of treatment to disease progression or death. * Overall Survival (OS): Time from the start of treatment until death.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of AEs from informed consent through 60 days following the last dose of AZD9829. Special monitoring for hematological toxicities and expected topoisomerase 1 inhibitor effects.
 - **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** Safety Review Committee (SRC) evaluations implemented to review DLTs and make dose-escalation cohort decisions.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all participants who receive ≥ 1 dose of AZD9829) and Efficacy Set (participants with measurable disease at baseline and post-baseline).
 - **Sample Size Justification:** Modular design scaling up to approximately 56 enrolled subjects based on standard Phase I/II dose-escalation and cohort-expansion statistical principles.
 - **Handling of Missing Data:** Standard oncology conventions applied; time-to-event outcomes (like OS and PFS) utilizing right-censoring methods for patients lost to follow-up or without events.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, focusing heavily on safety data in early dose cohorts.
 - **Audit Trail:** Centralized eCRF data collection, automated edit checks, and standard data clarification form processes.
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11. Study Identifiers & Governance

- **Primary ID:** D9470C00001
 - **Registry Number:** NCT06179511
 - **Sponsor/Lead Organization:** AstraZeneca
 - **Study Title:** Study of AZD9829 in CD123+ Hematological Malignancies
 - **Official Regulatory Title:** A Modular Phase I/II, Open-label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Antitumor Activity of AZD9829 as Monotherapy or in Combination in Patients With CD123-Positive Hematological Malignancies
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12. Clinical Framework (CD123 Malignancy Specific)

- **Primary Condition:** Relapsed/Refractory AML and High-Risk MDS
 - **Diagnosis Threshold:** CD123+ hematologic malignancy by flow cytometry/immunohistochemistry
 - **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) Delivery
 - **Optimization Target:** RP2D identification and leukemic blast reduction
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13. Study Procedures & Methodology

- **Management Pathway:** Dose-escalation and expansion oncology pathway
 - **Education Protocol (HCP):** Protocol and DLT monitoring training specific to AZD9829
 - **Education Protocol (Patient):** Onsite informed consent and safety/side-effect education
 - **Quality Audit Mechanism:** Safety Review Committee dose escalation evaluations
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14. Observation & Follow-Up Schedule

- **Total Duration:** Approximately 52 weeks (1 year)
 - **Onsite Visit Cadence:** Frequent monitoring during Cycle 1 (DLT evaluation window), then per regular treatment cycle
 - **Remote Monitoring Frequency:** Continuous safety monitoring through 60 days post-last dose
 - **Checklist Review Interval:** At each treatment cycle evaluation
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |